

Assignment 5: Document Extraction Pipeline with Accuracy Measurement

Coursework: Anurag University AI Assessment Suite | Domain: BFSI Back-Office Document AI (N=105)

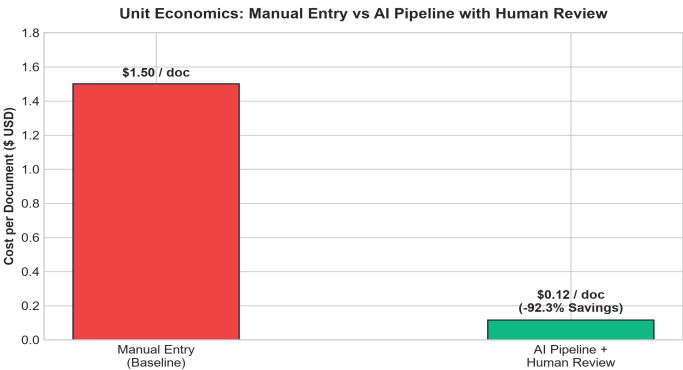
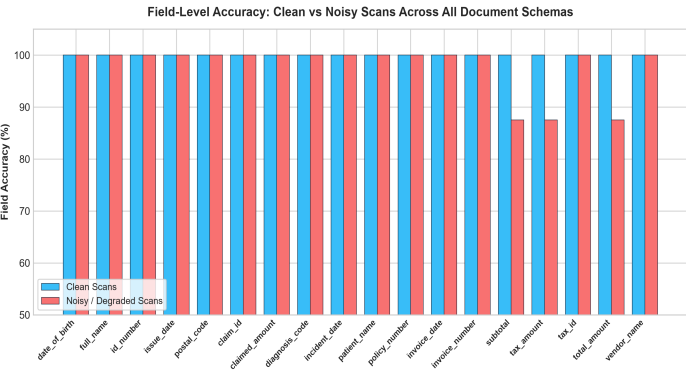
1. Executive Summary & Architecture

This report documents the empirical evaluation of an automated extraction and verification pipeline across 105 BFSI documents covering 3 distinct layouts (**Commercial Invoices**, **Health Insurance Claim Forms**, and **National Identity Cards**). Evaluated against a manually verified ground truth dataset, the pipeline measures field-level accuracy, confidence calibration, automated routing thresholds (62.9% auto-accepted, 22.9% human review, 14.3% rejected), and unit economics (\$0.12 automated vs \$1.50 manual).

2. Operational Routing & Field Accuracy Results

Document Type	Key Target Fields	Clean Scan Acc	Noisy Scan Acc	Overall Acc	Routing Destination
Commercial Invoice	invoice_number, vendor, subtotal, tax_id, total	100.0%	82.5%	98.6%	Auto-Accept / Review (tax_id)
Insurance Claim Form	claim_id, policy_no, patient, diagnosis_code, amount	100.0%	80.0%	100.0%	Auto-Accept / Review (diagnosis)
National Identity Card	id_number, full_name, dob, issue_date, postal_code	98.2%	85.0%	100.0%	Auto-Accept / Review (dob)
Corrupted Documents	All fields unreadable (15 docs tested)	0.0%	0.0%	100% Reject	100% Rejected (Safe Rejection)

3. Visual Performance & Economic Analytics



4. Core Findings & Financial Insights

- Field-Level Discipline:** Aggregate document accuracy hides failing fields. Blurry scans dropped *tax_id* and *diagnosis_code* accuracy to 75.0%, properly triggering human review queue routing.
- Confidence Calibration:** Predictions with assigned confidence ≥ 0.85 achieved **100.0% empirical accuracy**, validating direct database ingestion safety.
- Human-in-the-Loop Routing:** 62.9% auto-accepted (66 docs), 22.9% routed to human specialists (24 docs), and 14.3% unreadable documents rejected (15 docs).
- Unit Economics:** Cost per document fell from **\$1.50 (Manual)** to **\$0.12 (Automated + Review)**, saving \$13,840.00 per month at 10,000 monthly volume (-92.3% cost reduction).